Shifting Tornado Hazard in the United States and Emerging Insurance Exposure

Merrimack College
Data Science Capstone
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1 Background & Question

There are two major regions in the United States that are commonly associated with elevated tornado activity: the Southeast's region of "Dixie Alley", which includes Arkansas, Louisiana, Mississippi, Alabama, Tennessee and Georgia, and the Great Plains region of "Tornado Alley", which includes Texas, Oklahoma, Kansas and Nebraska. Recent research suggests that tornado activity may be shifting geographically and increasing in frequency in Dixie Alley while decreasing in frequency in Tornado Alley (Gensini and Brooks, 2018).

This capstone project asks if tornado hazard in the United States is shifting towards the Dixie Alley region over time, whether counties predicted to exhibit elevated or increasing tornado hazard also have high present-day insurance exposure based on population, housing units and median home value. It will also estimate the likelihood of high tornado hazard the following year. Aligning with the insurance industry, hazard refers to the frequency and severity of tornadoes and exposure refers to insurance-related exposure such as population, housing and home value.

This research question is motivated by the importance of understanding the changing trends in tornado risk through an insurance lens. If tornado activity is becoming more frequent and severe in areas with a higher housing and population density, like Dixie Alley, insurance companies may be faced with increased exposure and loss potential. The idea of this shift in tornado activity has been explored previously and is not a novel question, however, this project provides a predictive modeling framework to identify counties at risk of elevated tornado hazard and whether those counties also represent a greater present-day insurance exposure.

County-year is the selected unit of analysis for tornado hazard in this project because the research question has both a spatial and temporal component. It's valuable for stakeholders to not

only understand the counties with high tornado hazard, but also how that hazard is changing over time. Using county-year observations allows for deeper analysis into which counties are most frequently flagged as high hazard, which are becoming more hazardous over time, which are decreasing, and whether these patterns support a shift towards Dixie Alley potentially representing higher exposure for insurance companies.

1.1 Hypothesis and Prediction

Hypothesis: Tornado hazard is shifting from the traditional Tornado Alley towards Dixie Alley due to an increase in tornado frequency and severity in the Southeast over time.

Prediction: Counties located in Dixie Alley will be more likely be classified as higher tornado hazard in the following year compared to Tornado Alley. Counties predicted to be high hazard are expected to be characterized by a higher historical tornado frequency, greater average and maximum tornado intensity, and greater tornado-related property loss. The predicted high hazard counties are expected to also exhibit a higher present-day insurance exposure as measured by population, housing units and median home value.

2 Data & Methods

The historical tornado data used in this capstone project was obtained from the Severe Weather Database file (1950-2024) provided by the National Oceanic and Atmospheric Administration (NOAA) National Weather Service Storm Data (NOAA, 2024). The dataset was downloaded as a CSV file and includes detailed tornado records across the United States from 1950-2024 including the event date (year, month and day), location, tornado intensity (F-Scale or

EF-Scale) and estimated property loss. This dataset is appropriate for this project as it will allow for analysis of tornado hazard data including frequency and severity over a long portion of time.

County-level exposure data was obtained using the tidycensus package in R and the American Community Survey (ACS) 5-year estimates. Variables were selected that were well-suited for analyzing exposure relevant to insurance risk including, Estimate Total Population (B01003_001), Estimate Total Housing Units (B25001_001) and Estimate Median Value (Dollars) (B25077_001) for the year 2024. These variables allow for analysis of population, housing and home value exposure to evaluate insurance exposure.

Tornado hazard data spans the period of time between 1950-2024, while exposure variables are based on 2024 estimates only. Historical tornado data will be relevant for this dataset for identifying trends over time, however, present-day exposure data is significant to determine current insurance risk. This approach will allow for analysis of counties that have increased or increasing tornado hazard and pose a current-day insurance risk.

Variables were retrieved using the get_acs function and merged with the tornado dataset using county-level (Federal Information Processing Standards) FIPS identifiers to create a combined dataset that includes both historical tornado hazard and present-day exposure variables that will be valuable for analysis into the research question.

Model interpretability will be assessed using variable importance measures to ensure identification of key predictors of tornado hazard risk for stakeholders.

Lastly, an error analysis will also be considered for examination of misclassified observations, like false negatives, to better understand where the model may fail in the identification of high-risk counties.

3 Analysis Plan

The goal of this capstone is to predict whether the county-year observations will be classified as high tornado hazard in the following year using historical tornado hazard characteristics. Tornado hazard is measured by county-year levels of tornado frequency, intensity and property loss. In addition to identifying high tornado hazard county-years, the analysis will also evaluate whether the classifications have increased in frequency and severity in the Dixie Alley region over time.

An additional goal is to determine which counties show increasing tornado hazard over time and whether those counties also have a greater present-day insurance exposure. County-level exposure variables, including housing units, population and median home value will also be used to further evaluate areas where increased tornado hazard overlaps with present-day exposure for these variables.

These findings are intended to identify counties with elevated tornado hazard and evaluate where the hazard may overlap with greater insurance-related risk. This will allow stakeholders to identify where tornado hazard is increasing and whether those already also represent greater insurance exposure.

3.1.1 Data Preparation, Cleaning and Feature Engineering

Primary hazard analysis will be conducted at a county-year level and the present-day insurance exposure will be evaluated with county-level exposure variables. The tornado data needs to be prepared and cleaned to include the complete county Federal Information Processing Standard (FIPS) codes.

The Severe Weather Database file originally includes two separate columns for the state FIPS (stf) and county (f1-f4) FIPS columns, where f1 represents the originating county of the

tornado and f2-f4 represents any additional counties the tornado passed through. Leading zeroes were restored in where needed and state and county FIPS were combined to create the full standard five-digit FIPS code, called county_fips. For ease of interpretation, only the originating county (f1) was retained for analysis and others were omitted.

County-level exposure data obtained from The American Community Survey (ACS) included 2024 variables for: Estimate Total Population (B01003_001), Estimate Total Housing Units (B25001_001) and Estimate Median Value (Dollars) (B25077_001). These variables were selected due to the relevance for insurance exposure. Each ACS dataset was then cleaned for unique column headings and merged on the common identifier (GEOID), which is the full county-fips code. This was then renamed county_fips to allow merging with the tornado dataset.

The tornado dataset will be filtered to only include the years of 1979-2024 which aligns with the study period used by Gensini and Brooks (2018) and reflects more reliable modern tornado data reporting. Tornado hazard events will be aggregated to a county-year for hazard analysis.

A customized binary target variable, high_tornado_hazard_next_year, will be created to support predictive modeling. The variable will indicate whether a county is classified as high tornado hazard in the following year (1= high hazard and 0 = not high hazard).

The target variable is based on a composite tornado hazard score for county-year tornado frequency and severity: tornado_count_county_year, average_intensity_county, max_intensity_county, strong_tornado_count and total_property_loss_year. These will determine a hazard score where the top 25% will be indicated as (1) or high tornado hazard and anything else will be (0) or not high hazard. The target will be created by shifting the hazard classification

forward one year within each county allowing the model to predict future hazard conditions based on prior-year information.

Additional engineered features will be included for the following: region_category (T = Tornado Alley, D = Dixie Alley or O = Other), tornado_count_county_year, total_property_loss_year, average_property_loss, property_loss_per_tornado. As raw tornado counts may not accurately reflect present-day insurance exposure, normalized features will also be included: tornadoes_per_100k_population and tornadoes_per_10k_housing_units. This is due to the same tornado frequency may have differing insurance risk given vast differences in population and housing density. Additional details for each can be found in the Appendix in the itemized Data Dictionary.

Additional preprocessing is required for tornado intensity. The intensity scale for tornadoes changed in 2007 from the F-Scale (Fujita Scale) to the EF-Scale (Enhanced Fujita Scale) which measures the estimated tornado wind speeds based on damage left behind by a tornado (National Weather Service, NOAA, 2018). The major difference between these scales includes the addition to the EF-Scale for new damage indicators (i.e. building type, structures, trees, etc.) that were not incorporated in the original F-Scale (National Weather Service, NOAA, 2018). As the dataset includes data from 1979-2024, standardization between the F-Scale and EF-Scale is required. Though the wind speed ranges have changed between the F-Scale and EF-Scale, for the purposes of this capstone, the increase in intensity on either scale indicates a stronger tornado and will be standardized to a 0-5 scale rather than attempting to convert them based on estimated wind speeds. Additionally, anything coded as -9 (meaning unknown intensity) will be removed for intensity-related calculations.

F-Scale	EF-Scale	Standardized Scale

F0	EF0	0
F1	EF1	1
F2	EF2	2
F3	EF3	3
F4	EF4	4
F5	EF5	5

3.1.2 Data Analysis

Exploratory data analysis (EDA) will review spatial and temporal differences in tornado hazard across the United States. Initial analysis will include reviewing trends of county-year hazard variables including tornado counts, tornado intensity and property loss to identify trends and outliers. Present-day exposure variables, including population, housing and home value will also be analyzed for how exposure varies depending on tornado hazard. This general analysis overview will allow for understanding of initial patterns, outliers and potential relationships between each variable before predictive modeling using bar charts and box plots within R.

To evaluate whether tornado hazard is shifting towards Dixie Alley, analyzing regional trends in tornado frequency and intensity over time will be essential. Identifying which counties flag most frequently as high tornado hazard and which region they are located in, Dixie Alley, Tornado Alley or Other, will also be observed. Tableau will be utilized to prepare heat map visualizations of areas of high frequency of tornadoes and high tornado hazard county concentrations across the years. This analysis will help evaluate whether the data is consistent with prior findings suggesting a shift in tornado hazard from Tornado Alley to Dixie Alley.

It will also be reviewed to determine which counties are being flagged as high tornado hazard more frequently or with increasing frequency. This will allow determination of which counties are shifting to become a higher present-day insurance exposure and if they are more concentrated in Dixie Alley.

A limitation of this comparison is that the boundaries of Tornado Alley and Dixie Alley are difficult to establish as there are no universally agreed upon borders as they depend on the criteria being used (NOAA, 2026). For the purposes of this capstone project, Tornado Alley will be defined as Texas, Oklahoma, Kansas and Nebraska and Dixie Alley will be defined as Arkansas, Louisiana, Mississippi, Alabama, Tennessee and Georgia consistent with the groupings defined by Gagan, Gerard and Gordon (2010). Though this is a simplified approach, this will allow for a transparent boundary between the two regions.

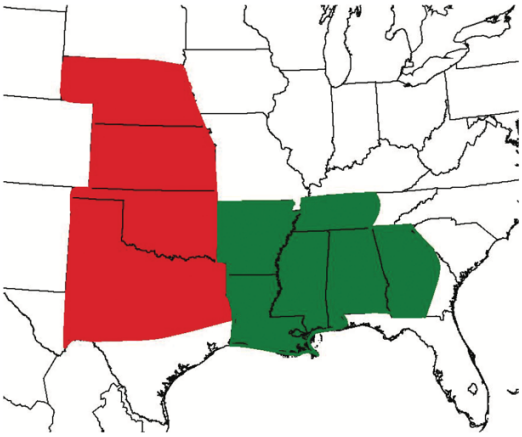

Fig. 1. A map outlining the Plains Tornado Alley, in red, and the Dixie Tornado Alley, in green.

Graph representation of significant Tornado Alley and Dixie Alley states from:
A Historical and Statistical Comparison of "Tornado Alley" to "Dixie Alley" (Gagan, Gerard & Gordon, 2010)

3.1.3 Predictive Modeling

As the data has a strong temporal structure, it will use a time-aware modeling approach. The data will be split using county-year observations, the training set will include the earlier years of 1979-2014 and the testing set will use the later years of 2015-2024. This will help ensure the model is evaluated on future data. However, it will also consider subgroup-aware cross-validation. This will ensure that when cross-validation is performed that it is subgroup-aware and observations for Tornado Alley and Dixie Alley are included. This will be perfomed with a stratified k-fold rather than a conventional k-fold. Hyperparameter tuning will be

conducted using cross-validation (randomized search) to optimize the model performance and reduce the risk of overfitting.

The target variable is high_tornado_hazard_next_year, which is a binary classification (1 = high tornado hazard, 0 = not a high tornado hazard). This will indicate whether a county will be classified as an elevated tornado hazard in the following year.

Predictor variables will include current-year historical tornado hazard characteristics and engineered features, which are used to predict whether a county will be classified as high tornado hazard in the following year. Regional indicators and county-level exposure variables will also be used to explore the relationship with next-year hazard classifications.

As the target variable is binary, logistic regression will be the appropriate starting model. It will be implemented in R using the glm function. Two additional machine learning models that will be utilized in this project include random forest and XGBoost as flexible models to analyze if performance improves when compared to the logistic regression model.

The models will be evaluated using recall and PR-AUC. Recall is prioritized as the primary evaluation metric to ensure identification of high-risk counties. Confusion matrices will provide a visualization of model performance.

The research question will be deemed answered if the model identifies counties that are likely to experience an increased tornado hazard in the following year, and if the predicted high hazard counties are increasingly concentrated in Dixie Alley and have a greater present-day insurance exposure.

4 Model Evaluation and Limitations

The primary coding language for this project will be R, and Tableau will be used for supplemental visualizations. The research question will be successfully answered if the model is able to distinguish between high tornado hazard and not high tornado hazard county-year observations and support whether tornado hazard is increasingly concentrated in Dixie Alley over time.

County-level exposure variables including, population, median home value and housing units and will be used to evaluate if an elevated tornado hazard corresponds to a higher county-level insurance exposure. This aligns best with the stakeholder's interests to be able to accurately determine the emerging high tornado hazard counties to support underwriting and risk management instead of looking at individual events.

The hypothesis will be considered supported if the county-year observations for Dixie Alley show a higher and increasing proportion of high tornado hazard classifications compared to Tornado Alley.

Several limitations should be acknowledged. The boundaries of Tornado Alley and Dixie Alley are not universally agreed upon. States were determined using the most commonly recognized states in each region and simplified for classification to include the entirety of each state. It was also substantiated by the article "A Historical and Statistical Comparison of 'Tornado Alley' to 'Dixie Alley'" by Gagan, Gerard & Gordon, 2010. Additionally, the track of the tornado was also simplified to include just the originating location of the tornado and did not include any subsequent counties that the tornado passed through. Lastly, the ACS data utilizes survey weights to balance disparities and the sample representative of the United States population. These weights are not incorporated into the machine learning model. Not using

weights does risk bias but this decision was made due to weights not being designed for predictive modeling.

As the data contains strong geographic structure, the model may overfit by learning spatial dependence across counties rather than relying on the intended variables and exposure variables. This will not be fully addressed in this capstone, however, future work could include incorporate spatial lagging of predictors to better account for geographic relationships.

The exposure variables used in this capstone are based on 2024 estimates only and it is not time-varying. While historical exposure data is not incorporated, population and housing data does gradually change over time making the use of 2024 estimates only appropriate.

5 GitHub

https://github.com/jill-baker/tornado-insurance-risk-repo

6 Appendix

The Data Dictionary below includes identification of key raw, engineered and target variables in the final county-year tornado hazard and insurance exposure dataset. The target variable, high_tornado_hazard_next_year, is created from the hazard-related variables in the dataset: tornado_count_county_year, average_intensity_county, max_intensity_county, total_property_loss_year and strong_tornado_count.

Column Name	Example	Data Type	Description
om	'192'	Integer	Tornado number. Count of tornadoes during given year from NOAA dataset

yr	'1950'	Integer	Predictor. Year of tornado event (1950-2024)
mo	'3'	Integer	Month of tornado event, (1-12)
dy	'26'	Integer	Day of tornado event, (1-31)
date	'1950-03-26'	Date	Date of tornado event in yyy-mm-dd format
st	'AR'	String	State (two-letter postal abbreviation) of tornado event
mag	'2'	Integer	F-Scale (EF-Scale after 01/07) Values 0, 1, 2, 3, 4, 5, -9 (-9=unknown)
loss	'4'	Numeric	Estimated property loss in millions of dollars. 0 or blank = unknown, not $0
len	'17.4'	Numeric	Length of tornado in miles
wid	'150'	Integer	Width of tornado in yards
county_fips	'05019'	String	Federal Information Processing Standards code for county, full 5 digit code (first 2 digits for state, last 3 digits for county).
county_name_state	'Clark County, Arkansas'	String	Full county name and state
population_estimate	'21125'	Integer	Predictor. Present-day exposure variable. Estimated total population in county
housing_unit_estimate	'10029'	Integer	Predictor. Present-day exposure variable. Estimated total number of housing units in the county
medvalue_estimate	'155000'	Numeric	Predictor. Present-day exposure variable. Estimated median home value in the county (in dollars)
region_category	'T'	String	Predictor. Indicates if county is located in Dixie Alley, Tornado Alley or Other. Dixie Alley = D, Tornado Alley = T, Other = O.
high_tornado_hazard_next_year	'1'	Binary	Target variable. Indicates whether a county is classified as high tornado hazard in the following year based on county's forward-shifted composite hazard classification 1=high tornado hazard, 0=not high tornado hazard
tornado_count_county_year	'4'	Integer	Target creation hazard variable. Total number of tornadoes per county for a given year
average_intensity_county	'3'	Integer	Target creation hazard variable. Average tornado intensity (F/EF-Scale) for tornadoes in county for a given year.
max_intensity_county	'5'	Integer	Target creation hazard variable. The maximum/highest observed tornado intensity per county in given year

total_property_loss_year	'5'	Integer	Target creation hazard variable. The total property loss with all occurred tornadoes per county in given year
average_property_loss	'4'	Numeric	Predictor. Hazard variable. The average property loss for tornadoes in county for a given year
strong_tornado_count	'3'	Integer	Target creation hazard variable. County of tornadoes in county-year where the tornado intensity (mag) is >=3
property_loss_per_tornado	'50000'	Numeric	Predictor. Calculation of total_property_loss_year / tornado_count_county_year
tornadoes_per_100k_population	'100'	Numeric	Predictor. Normalized measure: Calculation of tornado_count_county_year / population_estimate * 100000
tornadoes_per_10k_housing_units	'50'	Numeric	Predictor. Normalized measure: Calculation of tornado_count_county_year / housing_unit_estimate * 10000

7 References & Data Sets

- National Oceanic and Atmospheric Administration (NOAA), National Weather Service. (2024). *Severe Weather Database Files (1950–2024)*. Storm Prediction Center. https://www.spc.noaa.gov/wcm/
- U.S. Census Bureau. (2024). American Community Survey 5-Year Estimates. https://www.census.gov/programs-surveys/acs
  - Tidycensus in R with the following variables selected at the county-level: Estimate Total Population (B01003_001), Estimate Total Housing Units (B25001_001) and Estimate Median Value (Dollars) (B25077_001) for the year 2024
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